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Question Paper Code : 19300

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2022

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. - Information Technology

19MA303 – MATHEMATICS - III : DISCRETE MATHEMATICS

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

R, CO1 1. What is called tautology? Give an example.

U, CO1 2. State which rule of inference is used in the argument "If it rains today, the college will close. The college is not closed today. Therefore, it did not rain today". Justify your answer.

A) Modus Tollens B) Modus Ponens C) Addition D) Simplification

U, CO2 3. A group of 30 people have been trained as astronauts to go on the first mission to Mars. Then the number of ways to select a crew of six people to go on this mission is _____. Justify your answer.

A) 427518000 B) 720 C) 593775 D) None of these

R, CO2 4. State Pigeon hole principle.

U, CO3 5. The number of edges in a graph with 10 vertices each of degree 8 is _____. Justify your answer.

A) 80 B) 40 C) 18 D) None of the above

A, CO3 6. Find the chromatic number of complete graph K_4 and cycle graph C_5.

U, CO4 7. The set of all even integers$\{2,4,6,8,\ldots\}$ together with usual addition is a _____. Justify your answer.

A) Semigroup B) Monoid C) Group D) None of the above

R, CO4 8. Define field.

R, CO5 9. What is called Principle of duality?

U, CO5 10. In a Boolean Algebra, $(a.b) + (a.\bar{b}) =$_____. Justify your answer.

A) 0 B) 1 C) a D) None of the above

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. Verify whether the following $(P \longrightarrow Q) \wedge (R \longrightarrow S), (Q \longrightarrow T) \wedge (S \longrightarrow U), \neg(T \wedge U), P \longrightarrow R$ imply the conclusion $\neg P$. (8 Marks)

L,CO1 11. a) ii. Check the validity of the following arguments using predicate logic. "All integers are rational numbers", "Some integers are powers of 3", imply the conclusion "Some rational numbers are powers of 3". (8 Marks)

OR

A,CO1 11. b) i. Without using truth table, prove that

$p \rightarrow (q \vee r) \leftrightarrow \neg r \rightarrow (p \rightarrow q) \leftrightarrow (p \wedge \neg q) \rightarrow r$ (8 Marks)

L,CO1 11. b) ii. Verify that the following premises are inconsistent:

"If Rama gets his degree, he will go for a job", "If he goes for a job, he will earn money". "If he goes for higher study, he will not earn money", "Rama gets his degree and goes for higher study". (8 Marks)

A,CO2 12. a) i. Using the Principle of Mathematical induction to show that $8^n - 3^n$ is a multiple of 5. (8 Marks)

A,CO2 12. a) ii. Find all solutions of the recurrence relation

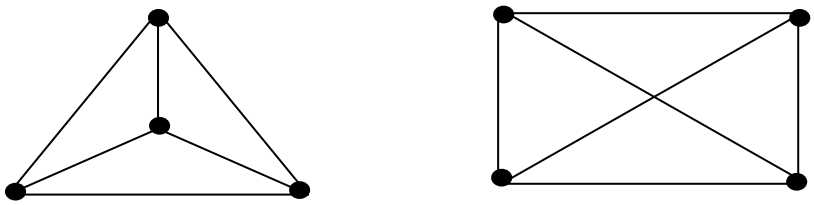
$a_n = 5a_{n-1} - 6a_{n-2} + 7^n$ (8 Marks)

OR

A,CO2 12. b) i. How many positive integers less than 1000 are divisible by 7, 11 or 13? (8 Marks)

A,CO2 12. b) ii. Suppose that a valid codeword is an n-digit number in decimal notation containing an even number of 0s. Let a_n denote the number of valid code words of length n. The sequence $\{a_n\}$ satisfies the recurrence relation $a_n = 8a_{n-1} + 10^{n-1}$, with the initial condition $a_0 = 1$. Find the solution using the method of generating functions. (8 Marks)

L,CO3 13. a) i. Verify whether the following graphs are isomorphic or not.

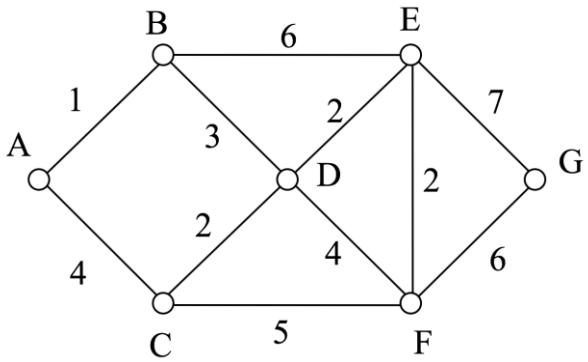


(8 Marks)

A,CO3 13. a) ii. Prove that an undirected graph has an even number of odd degree vertices. Check the above statement is true for the complete bipartite graph $K_{3,4}$ (8 Marks)

OR

L,CO3 13. b) i. Verify that the weight of the minimum spanning tree is 21 or 22 by Krushkal's algorithm.



(8 Marks)

A,CO3 13. b) ii. State and prove the necessary and sufficient condition for graph to be Eulerian. (8 Marks)

A,CO4 14. a) i. Show that the set of all positive rational numbers forms an abelian group under the composition defined by $a * b = \frac{ab}{2}$. (10 Marks)

A,CO4 14. a) ii. Prove that the intersection of two subgroups is a subgroup. (6 Marks)

OR

A,CO4 14. b) i. State and prove Lagrange's theorem with the necessary results required to prove it. (10 Marks)

A,CO4 14. b) ii. Show that if every element of a group G has its own inverse, then G is abelian. (6 Marks)

A,CO5 15. a) i. State and prove Modular inequality in a Lattice. (8 Marks)

A,CO5 15. a) ii. If $(L, \leq)$ is a lattice, then for any $a, b, c \in L$, prove the following property: If $b \leq c$, then A) $a \vee b \leq a \vee c$ B) $a \wedge b \leq a \wedge c$ (8 Marks)

OR

A,CO5 15. b) i. Prove that every chain is a distributive Lattice. (8 Marks)

A,CO5 15. b) ii. State and prove Demorgan's law in Boolean Algebra. (8 Marks)